Measure Theory

Notes

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1. Introduction – The Problem of Measure

Consider the set of all real numbers, $\mathbb{R}$. **Our goal is to assign a *measure* to every subset of $\mathbb{R}$.** Without defining this term rigorously, we have a certain intuitive notion of how this measure on $\mathbb{R}$ corresponds to length. For instance, it makes sense to say that the measure of the interval (a, b) is $b - a$, and that translating this interval by a certain quantity, say x, should not result in a change in its measure; that is, the measure of $(a + x, b + x)$ should remain $b - a$ for all $x \in \mathbb{R}$. We would like to extend this notion from open intervals like (a, b) to all subsets of $\mathbb{R}$.

Our objective, then, is to find a function λ that satisfies the following:

(P1) To every subset of $\mathbb{R}$, λ should assign some nonnegative value, possibly $+\infty$. Hence, it should be a function from $\mathscr{P}(\mathbb{R})$ to $[0, +\infty]$, i.e., $\lambda : \mathscr{P}(\mathbb{R}) \to [0, +\infty]$.

(P2) $\lambda((a, b)) = b - a$.

(P3) $\lambda(A + x) = \lambda(A)$, where $A + x := \{a + x : a \in A\}$.

(P4) Let $A_1, A_2, \ldots$ be a sequence of pairwise disjoint subsets of $\mathbb{R}$. Then, $\lambda(\bigsqcup_{j\geq 1} A_j) = \sum_{j\geq 1} \lambda(A_j)$.

We now show that such a function cannot exist.

Lemma 1.1. *Let E and F be subsets of $\mathbb{R}$ such that $E \subseteq F$. Then, $\lambda(E) \leq \lambda(F)$.*

Proof. We first show that $\lambda(\varnothing) = 0$. Set $A_j = \varnothing$ for all $j \geq 1$ in (P4). Then, $\lambda(\varnothing) = \lambda\left(\bigsqcup_{j\geq 1} \varnothing\right) = \sum_{j\geq 1} \lambda(\varnothing)$, which is possible only if $\lambda(\varnothing) = 0$.

If $E \subseteq F$, then $F = E \cup (F \setminus E)$, and E and $F \setminus E$ are disjoint. Setting $A_1 = E$, $A_2 = F \setminus E$, and $A_j = \varnothing$ for all $j \geq 3$ in (P4), we get $\lambda(F) = \lambda(E) + \lambda(F \setminus E) \geq \lambda(E)$. **QED**

Theorem 1.2. *A function λ satisfying (P1) to (P4) does not exist.*

Proof. Suppose, for contradiction, that such a function λ does exist.

Define a relation $\sim$ such that $a \sim b$ if and only if $a - b \in \mathbb{Q}$.

Clearly, $\sim$ is an equivalence relation. Define $[x] := \{y \in \mathbb{R} : x - y \in \mathbb{Q}\}$. Then, $[x]$ is an equivalence class. Observe that $[x] = \{x + q : q \in \mathbb{Q}\}$. Hence, there is a bijection $q \mapsto x + q$ from $\mathbb{Q}$ to $[x]$, and so $[x]$ is a countable set.

Let Λ be the collection of all equivalence classes. Then, Λ is uncountably infinite. For if Λ were to be countably infinite, then, since each equivalence class is also countable, the union of all the equivalence classes would be a countable union of countable sets, which would also be a countable set. However, the union of all equivalence classes is $\mathbb{R}$, since every real number belongs to some equivalence class, and $\mathbb{R}$ is uncountable, which is a contradiction.

By the Axiom of Choice, we can choose one element from each equivalence class. Call the resulting set Ω, and we may assume that $\Omega \subseteq (0,1)$, for we can choose an element from each equivalence class that is in $(0,1)$.

We first show that $\lambda(\Omega) = 0$.

Let $q_1, q_2 \in \mathbb{Q}$, $q_1 \neq q_2$. We now show that $(\Omega + q_1) \cap (\Omega + q_2) = \varnothing$.

Let $x \in (\Omega + q_1) \cap (\Omega + q_2)$. Then, $x \in \Omega + q_1$ and $x \in \Omega + q_2$, that is, $x = \alpha + q_1$ and $x = \beta + q_2$ for some $\alpha, \beta \in \Omega$. Hence, $\alpha + q_1 = \beta + q_2$, or, $\alpha - \beta = q_2 - q_1 \in \mathbb{Q}$. Therefore, $\alpha \sim \beta$, and so α and β belong to the same equivalence classes. However, both α and β are in Ω, which contains exactly one element from each equivalence class, by construction. This implies that $\alpha = \beta$, and so $q_1 = q_2$, which contradicts that $q_1 \neq q_2$, proving that $(\Omega + q_1) \cap (\Omega + q_2) = \varnothing$.

Now consider $-1 < q < 1$, $q \in \mathbb{Q}$. Since $\Omega \subseteq (0,1)$,

$$\bigsqcup_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} (\Omega + q) \subseteq (-1, 2),$$

and by (P4) along with the lemma,

$$\sum_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} \lambda(\Omega + q) \leq \lambda((-1,2)) = 3.$$

But by (P3), $\lambda(\Omega + q) = \lambda(\Omega)$, and so

$$\sum_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} \lambda(\Omega) \leq \lambda((-1,2)) = 3,$$

which is only possible if $\lambda(\Omega) = 0$, since there are infinitely many rationals between -1 and 1. Hence, $\lambda(\Omega) = 0$.

We now show:

$$(0,1) \subseteq \bigsqcup_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} (\Omega + q).$$

Suppose $x \in (0,1)$. Let $x_0 = [x] \cap \Omega$. Then, $x \sim x_0$, i.e., $x - x_0 = q \in \mathbb{Q}$. Now, $x_0 \in (0,1)$ (for $x_0 \in \Omega \subseteq (0,1)$), and $x \in (0,1)$, which implies that $q = x - x_0 \in (-1,1)$. Thus, $x = x_0 + q \in \Omega + q$ for some rational q such that $-1 < q < 1$. Hence, $x \in \bigsqcup_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} (\Omega + q)$.

By the lemma,

$$\lambda((0,1)) \leq \lambda\left(\bigsqcup_{\substack{-1<q<1 \\ q\in\mathbb{Q}}} (\Omega + q) \right),$$

i.e.,

$$1 \leq \sum_{\substack{-1<q<1 \\ q \in \mathbb{Q}}} \lambda(\Omega + q),$$

which is a contradiction, for the sum on the right is 0. **QED**

Thus, we must accept that, with the Axiom of Choice, it is not possible to associate such a "measure" satisfying (P1) - (P4) to all possible subsets of $\mathbb{R}$. Instead, we must try to find the "largest" class of subsets of $\mathbb{R}$ for which it is possible to assign such a measure.

2. Classes of Subsets

Let Ω be a set and let $\mathscr{P}(\Omega)$ denote the set of all subsets of Ω.

Definition 2.1. *Let $\mathscr{S}$ be a subset of $\mathscr{P}(\Omega)$. $\mathscr{S}$ is a* ***semi-algebra*** *if:*

1. $\Omega \in \mathscr{S}$;
2. *if $A, B \in \mathscr{S}$, then $A \cap B \in \mathscr{S}$; and,*
3. *if $A \in \mathscr{S}$, then there exist some $A_1, A_2, ..., A_n \in \mathscr{S}$, where $A_i \cap A_j = \varnothing$ if $i \neq j$ and $A^c = \bigsqcup_{i=1}^{n} A_i$.*

Example 2.2. Let $\Omega = \mathbb{R}$ and $\mathscr{S} = \{\varnothing, \mathbb{R}, (a, b], (-\infty, b], (a, \infty) : a, b \in \mathbb{R} \text{ and } a < b\}$. $\mathscr{S}$ is a semi-algebra.

Definition 2.3. *Let $\mathscr{A}$ be a subset of $\mathscr{P}(\Omega)$. $\mathscr{A}$ is an* ***algebra*** *if:*

1. $\Omega \in \mathscr{A}$;
2. *if $A, B \in \mathscr{A}$, then $A \cap B \in \mathscr{A}$; and,*
3. *if $A \in \mathscr{A}$, then $A^c \in \mathscr{A}$.*

Note that this definition implies that an algebra is also closed under finite unions. That is, if $A, B \in \mathscr{A}$, then $A \cup B \in \mathscr{A}$; for $A, B \in \mathscr{A}$ implies $A^c, B^c \in \mathscr{A}$, and by closure under intersection we have $A^c \cap B^c \in \mathscr{A}$, i.e., $(A \cup B)^c \in \mathscr{A}$. Hence, $A \cup B \in \mathscr{A}$.

Definition 2.4. *Let $\mathscr{F}$ be a subset of $\mathscr{P}(\Omega)$. $\mathscr{F}$ is a* ***σ-algebra*** *if:*

1. $\Omega \in \mathscr{A}$;
2. *if $A_j \in \mathscr{F}$ for all $j \geq 1$, then $\bigcup_{j \geq 1} A_j \in \mathscr{F}$; and,*
3. *if $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$.*

Again, this definition of a σ-algebra implies closure under countable intersections as well.

Observe that:

- Every algebra is a semi-algebra.
- Every σ-algebra is an algebra.

Remark 2.5. 1. Let $\mathscr{A}_\alpha \subseteq \mathscr{P}(\Omega)$, $\alpha \in I$, be a family of algebras, where I is an index set, possibly uncountable. Then, $\bigcap_{\alpha\in I} \mathscr{A}_\alpha$ is also an algebra.

2. Similarly, if $\mathscr{F}_\alpha \subseteq \mathscr{P}(\Omega)$, $\alpha \in I$ is a family of σ-algebras, where I is a possibly uncountable index set, then, $\bigcap_{\alpha\in I} \mathscr{F}_\alpha$ is also a σ-algebra.

Definition 2.6. *Let $\mathscr{C} \subseteq \mathscr{P}(\Omega)$. Then, an algebra $\mathscr{A}(\mathscr{C})$ is the **algebra generated by** $\mathscr{C}$ if $\mathscr{C} \subseteq \mathscr{A}(\mathscr{C})$ and $\mathscr{A}(\mathscr{C}) \subseteq \mathscr{B}$ for any algebra $\mathscr{B}$ containing $\mathscr{C}$ (i.e., $\mathscr{C} \subseteq \mathscr{B}$).*

$\mathscr{A}(\mathscr{C})$ is the "smallest" algebra containing $\mathscr{C}$. Such an algebra exists, for $\mathscr{P}(\Omega)$ is an algebra which contains $\mathscr{C}$, and by Remark 2.5, the intersection of all algebras containing $\mathscr{C}$ is also an algebra. This is precisely the algebra generated by $\mathscr{C}$.

Definition 2.7. *Let $\mathscr{C} \subseteq \mathscr{P}(\Omega)$. Then, a σ-algebra $\sigma(\mathscr{C})$ is the **σ-algebra generated by** $\mathscr{C}$ if $\mathscr{C} \subseteq \sigma(\mathscr{C})$ and $\sigma(\mathscr{C}) \subseteq \mathscr{F}$ for any σ-algebra $\mathscr{F}$ containing $\mathscr{C}$ (i.e., $\mathscr{C} \subseteq \mathscr{F}$).*

Again, $\sigma(\mathscr{C})$ exists, for $\mathscr{P}(\Omega)$ is a σ-algebra containing $\mathscr{C}$ and by Remark 2.5, the intersection of all σ-algebras containing $\mathscr{C}$ is also a σ-algebra, which is in fact the σ-algebra generated by $\mathscr{C}$.

Lemma 2.8. *Let $\mathscr{S} \subseteq \mathscr{P}(\Omega)$ be a semi-algebra. The algebra generated by $\mathscr{S}$, $\mathscr{A}(\mathscr{S})$, is the collection of all subsets of Ω that can be expressed as a finite disjoint union of sets in $\mathscr{S}$. That is, $A \in \mathscr{A}(\mathscr{S})$ if and only if there exist disjoint $E_j \in \mathscr{S}$, $1 \le j \le n$, such that $A = \bigsqcup_{j=1}^n E_j$.*

Proof. We first show that if $A = \bigsqcup_{j=1}^n E_j$ for some disjoint $E_j \in \mathscr{S}$, then $A \in \mathscr{A}(\mathscr{S})$. Since $E_j \in \mathscr{S}$ and $\mathscr{S} \subseteq \mathscr{A}(\mathscr{S})$, $E_j \in \mathscr{A}(\mathscr{S})$. $\mathscr{A}(\mathscr{S})$ is an algebra and so $\bigsqcup_{j=1}^n E_j \in \mathscr{A}(\mathscr{S})$.

We now show the reverse direction. Define

$$\mathscr{B} = \left\{ \bigsqcup_{j=1}^n F_j : F_j \in \mathscr{S} \right\}.$$

Then, $\mathscr{B}$ is the collection of all subsets of Ω which are finite disjoint unions of sets in $\mathscr{S}$. We show that $\mathscr{B}$ is an algebra.

Clearly, $\mathscr{S} \subseteq \mathscr{B}$. Since $\Omega \in \mathscr{S}$, $\Omega \in \mathscr{B}$.

Closure under finite intersections. Let $A, B \in \mathscr{B}$. Then, $A = \bigsqcup_{j=1}^n E_j$ and $B = \bigsqcup_{k=1}^m F_k$ for some $E_j, F_k \in \mathscr{S}$. Now,

$$A \cap B = \left(\bigsqcup_{j=1}^n E_j \right) \cap \left(\bigsqcup_{k=1}^m F_k \right) = \bigsqcup_{j=1}^n \bigsqcup_{k=1}^m (E_j \cap F_k)$$

where the second equality is due to the distributive property: $\left(\bigsqcup_{j=1}^n E_j\right) \cap (\bigsqcup_{k=1}^m F_k)$ is the disjoint union over all pairs $(E_j \cap F_k)$. It is a disjoint union because $(E_{j_1} \cap F_{k_1}) \cap (E_{j_2} \cap F_{k_2}) = \varnothing$, as E_{j_1}, E_{j_2} are disjoint and F_{k_1}, F_{k_2} are disjoint. Hence, $A \cap B$ is a finite disjoint union of sets in $\mathscr{S}$ and so $A \cap B \in \mathscr{B}$, i.e., $\mathscr{B}$ is closed under finite intersections.

Closure under complementation. Now suppose $A \in \mathscr{B}$. Then, $A = \bigsqcup_{j=1}^{n} E_j$ where $E_j \in \mathscr{S}$. Therefore,

$$A^c = \left(\bigsqcup_{j=1}^{n} E_j \right)^c = \bigcap_{j=1}^{n} E_j^c = E_1^c \cap ... \cap E_n^c.$$

Since $E_j \in \mathscr{S}$, $E_j^c = \bigsqcup_{k_i=1}^{l_i} F_{i,k_i}$ for some $F_{i,j} \in \mathscr{S}$. Hence,

$$\begin{aligned} A^c &= \left(\bigsqcup_{k_1=1}^{l_1} F_{1,k_1} \right) \cap ... \cap \left(\bigsqcup_{k_n=1}^{l_n} F_{n,k_n} \right) \\ &= \bigsqcup_{k_1=1}^{l_1} ... \bigsqcup_{k_n=1}^{l_n} (F_{1,k_1} \cap ... \cap F_{n,k_n}) \end{aligned}$$

by the distributive property. Again, $F_{1,k_1} \cap ... \cap F_{n,k_n} \in \mathscr{S}$ and they are pairwise disjoint. Thus, A^c is a finite disjoint union of sets in $\mathscr{S}$, i.e., $A^c \in \mathscr{B}$.

We have shown that $\mathscr{B}$ is an algebra. Since $\mathscr{S} \subseteq \mathscr{B}$, $\mathscr{A}(\mathscr{S}) \subseteq \mathscr{B}$. Hence, if $A \in \mathscr{A}(\mathscr{S})$, then $A \in \mathscr{B}$ and therefore A can be expressed as a finite disjoint union of sets in $\mathscr{S}$. **QED**

3. Measure

Let Ω be a set.

Definition 3.1. *Let $\mathscr{F} \subseteq \mathscr{P}(\Omega)$ be a σ-algebra, and let $\mu : \mathscr{F} \to [0, +\infty]$ be a function such that:*

(1) $\mu(\varnothing) = 0$

*(2) (**Countable additivity** or **σ-additivity**.) If $(E_j)_{j\geq 1}$ is a sequence of of pairwise disjoint sets in $\mathscr{F}$, then*

$$\mu\left(\bigsqcup_{j\geq 1} E_j\right) = \sum_{j\geq 1} \mu(E_j).$$

*Then, μ is called a **measure**.*

Countable additivity implies **finite additivity**: If $E_1, ..., E_n$ is a sequence of pairwise disjoint sets in $\mathscr{F}$, then

$$\mu\left(\bigsqcup_{j=1}^{n} E_j\right) = \sum_{j=1}^{n} \mu(E_j).$$

Proposition 3.2. *Let $E, F \in \mathscr{F}$ such that $E \subseteq F$ and let μ be a measure on $\mathscr{F}$. Then, $\mu(E) \leq \mu(F)$. Further, if $\mu(E) < +\infty$, then $\mu(F \setminus E) = \mu(F) - \mu(E)$.*

Proof. Since $E \subseteq F$, we have $F = (E) \sqcup (F \setminus E)$, and so $\mu(F) = \mu(E) + \mu(F \setminus E)$. Since $\mu(F \setminus E)$ is nonnegative, $\mu(F) \geq \mu(E)$.

Now, let $\mu(E) < +\infty$. Since $\mu(F) = \mu(E) + \mu(F \setminus E)$ and $\mu(E) < +\infty$, we can subtract $\mu(E)$ from both sides to obtain $\mu(F \setminus E) = \mu(F) - \mu(E)$. **QED**

Example 3.3. (**Discrete measure.**) Let $\mathscr{F} \subseteq \mathscr{P}(\Omega)$ be a σ-algebra. Let $x_1, x_2, ... \in \Omega$ and $p_1, p_2, ... \geq 0$. For all $A \in \mathscr{F}$, define

$$\mu(A) := \sum_{j\geq 1} p_j \mathbf{1}_A(x_j)$$

where $\mathbf{1}$ is the indicator function:

$$\mathbf{1}_A(x_j) = \begin{cases} 1, \text{ if } x_j \in A \\ 0, \text{ if } x_j \notin A \end{cases}.$$

Then, μ is a measure.

Example 3.4. Let $\Omega = (0,1)$ and $\mathscr{C} := \{(a,b] : 0 \leq a < b < 1\}$. Define $\mu : \mathscr{C} \to [0,+\infty]$ by

$$\mu((a,b]) = \begin{cases} +\infty, & \text{if } a = 0 \\ b-a, & \text{if } a > 0 \end{cases}.$$

Then, μ is finitely additive, but not countably additive. To see this, consider a strictly monotonic decreasing sequence $x_n \to 0$ where $x_1 = \frac{1}{2}$. Then,

$$\bigsqcup_{j\geq 1}(x_{j+1}, x_j] = \left(0, \frac{1}{2}\right]$$

and so $\mu\left(\bigsqcup_{j\geq 1}(x_{j+1}, x_j]\right) = 0$, but $\sum_{j\geq 1}\mu((x_{j+1}, x_j]) = \sum_{j\geq 1}(x_j - x_{j+1}) = (x_1 - x_2) + (x_2 - x_3) + \ldots = \frac{1}{2}$.

Lemma 3.5. *Let μ be a measure on a σ-algebra $\mathscr{F}$ and $E_n, E \in \mathscr{F}$ for all $n \geq 1$.*
*(1) (**Continuity from below.**) If $E_n \uparrow E$, then $\lim_{n\to\infty}\mu(E_n) = \mu(E)$.*
*(2) (**Continuity from above.**) If $E_n \downarrow E$ and $\mu(E_N) < +\infty$ for some $N \in \mathbb{N}$, then $\lim_{n\to\infty}\mu(E_n) = \mu(E)$.*

Proof. (1) We have $E_1 \subseteq E_2 \subseteq \ldots \subseteq E$ and $\bigcup_{j\geq 1} E_j = E$. Define $F_1 := E_1$ and $F_j := E_j \backslash E_{j-1}$ for $j \geq 2$. Then, $(F_j)_{j\geq 1}$ is a disjoint sequence of sets. Further,

$$\bigsqcup_{j\geq 1} F_j = \bigcup_{j\geq 1} E_j = E$$

and so

$$\mu\left(\bigsqcup_{j\geq 1} F_j\right) = \mu(E).$$

Since μ is a measure,

$$\sum_{j\geq 1}\mu(F_j) = \mu(E). \tag{*}$$

Observe that $\bigsqcup_{j=1}^{n} F_j = F_1 \sqcup F_2 \sqcup \ldots \sqcup F_n = (E_1) \sqcup (E_2 \setminus E_1) \sqcup \ldots \sqcup (E_n \setminus E_{n-1}) = E_n$ and so

$$\mu\left(\bigsqcup_{j=1}^{n} F_j\right) = \mu(E_n),$$

that is,

$$\sum_{j=1}^{n}\mu(F_j) = \mu(E_n).$$

Taking the limit as $n \to \infty$ on both sides,

$$\sum_{j\geq 1}\mu(F_j) = \lim_{n\to\infty}\mu(E_n).$$

Combining this with (*) gives $\lim_{n\to\infty}\mu(E_n)=\mu(E)$, as desired.

(2) We have $E_1\supseteq E_2\supseteq\ldots\supseteq E$ and $\bigcap_{j\geq 1}E_j=E$. Define $G_1:=E_N\setminus E_{N+1}$, $G_2:=E_N\setminus E_{N+2}$, and so on.

Since $E_n\downarrow E$, we have $E_{N+n}\downarrow E$, and so $(E_N\setminus E_{N+n})\uparrow(E_N\setminus E)$, i.e., $G_n\uparrow(E_N\setminus E)$. By (1), $\lim_{n\to\infty}\mu(G_n)=\mu(E_N\setminus E)$. Since $E\subseteq E_N$, by Proposition 3.2 we have $\mu(E)\leq\mu(E_N)$. But $\mu(E_N)<+\infty$ and so $\mu(E)<+\infty$. By Proposition 3.2, we get that $\mu(E_N\setminus E)=\mu(E_N)-\mu(E)$. Similarly, $\mu(G_n)=\mu(E_N\setminus E_{N+n})=\mu(E_N)-\mu(E_{N+n})$. Hence,

$$\lim_{n\to\infty}\mu(E_N)-\mu(E_{N+n})=\mu(E_N)-\mu(E)$$

and so $\lim_{n\to\infty}\mu(E_{N+n})=\mu(E)$, i.e., $\lim_{n\to\infty}\mu(E_n)=\mu(E)$. **QED**

Lemma 3.6. *Let*

4. Carathéodory's Extension Theorem

Definition 4.1. *(**Outer measure.**) Let Ω be a set. Then, $\mu^* : \mathscr{P}(\Omega) \to [0, \infty]$ is called an outer measure on Ω if:*

1. $\mu^*(\varnothing) = 0$.
2. *(**Monotonicity.**) If $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$.*
3. *(**Sub-additivity.**) If $E \subseteq \bigcup_{j\geq 1} E_j$, then $\mu^*(E) \leq \sum_{j\geq 1} \mu^*(E_j)$.*

Proposition 4.2. *Let $\mathscr{C}$ be a collection of subsets of Ω such that $\varnothing, \Omega \in \mathscr{C}$. Let $\rho : \mathscr{C} \to [0, \infty]$ such that $\rho(\varnothing) = 0$. Define $\mu^* : \mathscr{P}(\Omega) \to [0, \infty]$ by*

$$\mu^*(A) := \inf\left\{\sum_{j\geq 1} \rho(A_j) : A \subseteq \bigcup_{j\geq 1} A_j \text{ and } A_j \in \mathscr{C} \ \forall\ j \geq 1\right\}.$$

Then, μ^* is an outer measure on Ω.

Proof. We first note that since $\Omega \in \mathscr{C}$, for any $A \in \mathscr{P}(\Omega)$, $A \subseteq \Omega$ and so we can set $A_1 = \Omega$ and $A_j = \varnothing$ for all $j \geq 2$. Then, $A \subseteq \bigcup_{j\geq 1} A_j$ and hence μ^* is well-defined.

We show that it is an outer measure as follows:

1. We start by showing that $\mu^*(\varnothing) = 0$. Set $A_j = \varnothing$ for all $j \geq 1$. Then, $\varnothing \subseteq \bigcup_{j\geq 1} A_j$ and so $\mu^*(\varnothing) \leq \sum_{j\geq 1} \rho(A_j) = 0$ since $\rho(\varnothing) = 0$. Further, if $\{A_j\}_{j\geq 1}$ is any covering of $\varnothing$, then $\sum_{j\geq 1} \rho(A_j) \geq 0 \implies \mu^*(\varnothing) \geq 0$. Hence, $\mu^*(\varnothing) = 0$.

2. We now show monotonicity: let $A \subseteq B$. We want to show that $\mu^*(A) \leq \mu^*(B)$. Let $\{B_j\}_{j\geq 1}$ be a covering of B. Then, $B \subseteq \bigcup_{j\geq 1} B_j$, but since $A \subseteq B$, we have that $A \subseteq \bigcup_{j\geq 1} B_j$, that is, $\{B_j\}_{j\geq 1}$ is also a covering of A. Thus, every covering of B is a covering of A, which implies that $\mu^*(A)$ is the infimum taken over a set which contains the set over which the infimum is taken for $\mu^*(B)$. Hence, $\mu^*(A) \leq \mu^*(B)$.

3. Finally, we show sub-additivity: if $A \subseteq \bigcup_{j\geq 1} A_j$, then $\mu^*(A) \leq \sum_{j\geq 1} \mu^*(A_j)$. We may assume that $\mu^*(A_j) < \infty$ for all $j \geq 1$, for if that is not the case then the assertion is trivial.

   By definition of μ^*, for all $\varepsilon > 0$, there exists some covering $\{E_{j,k}\}_{k\geq 1}$ of A_j in $\mathscr{C}$ such that

$$\mu^*(A_j) \leq \sum_{k\geq 1} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\varepsilon}{2^j}.$$

Now, $A \subseteq \bigcup_{j\geq 1} A_j \subseteq \bigcup_{j,k\geq 1} E_{j,k}$, and therefore,

$$\mu^*(A) \leq \sum_{j,k\geq 1} \rho(E_{j,k}) \leq \sum_{j\geq 1} \left(\mu^*(A_j) + \frac{\varepsilon}{2^j}\right) = \sum_{j\geq 1} \mu^*(A_j) + \varepsilon.$$

Since this is true for all $\varepsilon > 0$, we obtain $\mu^*(A) \leq \sum_{j\geq 1} \mu^*(A_j)$, as required.

Hence, μ^* is an outer measure on Ω. **QED**

Definition 4.3. *Let μ^* be an outer measure on Ω. Then, $A \subseteq \Omega$ is said to be μ^*-measurable or Carathéodory-measurable if*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for all $E \subseteq \Omega$.

Theorem 4.4. ***(Carathéodory's Extension Theorem.)*** *Let μ^* be an outer measure on Ω and let $\mathscr{M}$ be the collection of all μ^*-measurable sets. Then:*

1. *$\mathscr{M}$ is a σ-algebra.*
2. *The restriction of μ^* on $\mathscr{M}$, $\mu^*|_{\mathscr{M}}$, is a complete measure.*

Proof. 1. We first show that $\mathscr{M}$ is a σ-algebra.

Note that for any $E \subseteq \Omega$, $E \subseteq (E \cap A) \cup (E \cap A^c)$ and therefore $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ due to the sub-additivity of μ^*. Hence, to show that $A \in \mathscr{M}$, it suffices to show that $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Since this is trivial if $\mu^*(E) = \infty$, we consider the case where $\mu^*(E) < \infty$.

(a) Since $\mu^*(E \cap \varnothing) + \mu^*(E \cap \varnothing^c) = \mu^*(\varnothing) + \mu^*(E) = 0 + \mu^*(E) = \mu^*(E)$ for all $E \subseteq \Omega$, $\varnothing \in \mathscr{M}$.

(b) $\mathscr{M}$ is closed under complements since the definition of μ^* -measurable sets is symmetric about complements: if $A \in \mathscr{M}$, then for all $E \subseteq \Omega$,

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A^c) + \mu^*(E \cap A) \\ &= \mu^*(E \cap A^c) + \mu^*(E \cap (A^c)^c)\end{aligned}$$

and therefore $A^c \in \mathscr{M}$.

(c) It remains to show that $\mathscr{M}$ is closed under countable unions. If $A_j \in \mathscr{M}$ for all $j \geq 1$, we want to show that $\bigcup_{j\geq 1} A_j \in \mathscr{M}$.

In order to show this, we first prove closure under finite unions. Let $A, B \in \mathscr{M}$. We want to show that $A \cup B \in \mathscr{M}$, that is,

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

for all $E \subseteq \Omega$. Since $A \in \mathscr{M}$,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Further, since $B \in \mathscr{M}$,

$$\mu^*(E \cap A^c) \geq \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

and so,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).$$

Equivalently,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap (A \cup B)^c). \tag{1}$$

But since $A \cup B = A \cup (B \setminus A) = A \cup (B \cap A^c)$,

$$E \cap (A \cup B) = E \cap (A \cup (B \cap A^c)) = (E \cap A) \cup (E \cap A^c \cap B).$$

Due to the sub-additivity of μ^*,

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B).$$

Using this in (1) yields $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since this is true for all $E \subseteq \Omega$, $A \cup B \in \mathscr{M}$. We have thus shown closure under finite unions, implying that $\mathscr{M}$ is an algebra.

We now show closure under countable unions. Let $A = \bigcup_{j\geq 1} A_j$. We want to show:

$$\mu^*(E) \geq \mu^* \left(E \cap \bigcup_{j\geq 1} A_j \right) + \mu^* \left(E \setminus \bigcup_{j\geq 1} A_j \right).$$

Due to closure under finite unions, we know:

$$\mu^*(E) \geq \mu^* \left(E \cap \bigcup_{j=1}^{n} A_j \right) + \mu^* \left(E \setminus \bigcup_{j=1}^{n} A_j \right).$$

Since $\bigcup_{j=1}^{n} A_j \subseteq \bigcup_{j\geq 1} A_j$, $\mu^*(E \setminus \bigcup_{j=1}^{n} A_j) \geq \mu^*(E \setminus \bigcup_{j\geq 1} A_j)$. Hence,

$$\mu^*(E) \geq \mu^* \left(E \cap \bigcup_{j=1}^{n} A_j \right) + \mu^* \left(E \setminus \bigcup_{j\geq 1} A_j \right). \tag{2}$$

Define $B_1 = A_1$, $B_2 = A_2 \setminus A_1$, and similarly, $B_j = A_j \setminus \bigcup_{k=1}^{j-1} A_k$. Clearly, B_j are pairwise disjoint and $B_j \in \mathscr{M}$ for all $j \geq 1$ because $\mathscr{M}$ is an algebra and thus is closed under set difference. Further, $\bigsqcup_{j\geq 1} B_j = \bigcup_{j\geq 1} A_j$.

We now show:

$$\mu^*\left(E \cap \bigsqcup_{j=1}^{n} B_j\right) = \sum_{j=1}^{n} \mu^*(E \cap B_j). \tag{3}$$

We show this by induction. The statement is trivial for $n = 1$. Suppose it is true for n. For $n+1$, since $B_{j+1} \in \mathscr{M}$, we have:

$$\begin{aligned}
\mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j\right) &= \mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j \cap B_{j+1}\right) + \mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j \cap B_{j+1}^c\right) \\
&= \mu^*(E \cap B_{j+1}) + \mu^*\left(E \cap \bigsqcup_{j=1}^{n} B_j\right) \\
&= \mu^*(E \cap B_{j+1}) + \sum_{j=1}^{n} \mu^*(E \cap B_j) \\
&= \sum_{j=1}^{n+1} \mu^*(E \cap B_j)
\end{aligned}$$

Hence, $\mu^*\left(E \cap \bigcup_{j=1}^{n} A_j\right) = \mu^*\left(E \cap \bigsqcup_{j=1}^{n} B_j\right) = \sum_{j=1}^{n} \mu^*(E \cap B_j)$. Using this in (2),

$$\mu^*(E) \geq \sum_{j=1}^{n} \mu^*(E \cap B_j) + \mu^*\left(E \setminus \bigcup_{j\geq 1} A_j\right)$$

This is true for all n. Hence,

$$\begin{aligned}
\mu^*(E) &\geq \sum_{j\geq 1} \mu^*(E \cap B_j) + \mu^*\left(E \setminus \bigcup_{j\geq 1} A_j\right) \\
&\geq \mu^*\left(\bigsqcup_{j\geq 1}(E \cap B_j)\right) + \mu^*\left(E \setminus \bigcup_{j\geq 1} A_j\right) && \text{(by subadditivity)} \\
&= \mu^*\left(E \cap \bigcup_{j\geq 1} A_j\right) + \mu^*\left(E \setminus \bigcup_{j\geq 1} A_j\right)
\end{aligned}$$

thereby showing that $\bigcup_{j\geq 1} A_j$ is in $\mathscr{M}$. Therefore, $\mathscr{M}$ is a σ-algebra.

2. We now show that $\mu^*|_{\mathscr{M}}$ is a complete measure.

   We begin by showing that it is a measure. Since $\mathscr{M}$ is a σ-algebra, $\varnothing \in \mathscr{M}$, and $\mu^*(\varnothing) = 0$.

We further need to show countable additivity. Let $\{A_j\}_{j\geq 1}$ be a sequence of disjoint sets in $\mathscr{M}$. We want to show that $\mu^*\left(\bigsqcup_{j\geq 1} A_j\right) = \sum_{j\geq 1}\mu^*(A_j)$.

By substituting $E = \Omega$ in (3), we obtain finite additivity:

$$\mu^*\left(\bigsqcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n}\mu^*(A_j). \tag{4}$$

By sub-additivity,

$$\mu^*\left(\bigsqcup_{j\geq 1} A_j\right) \leq \sum_{j\geq 1}\mu^*(A_j). \tag{5}$$

Also,

$$\bigsqcup_{j\geq 1} A_j \supseteq \bigsqcup_{j=1}^{n} A_j$$

and so, by monotonicity, we obtain

$$\mu^*\left(\bigsqcup_{j\geq 1} A_j\right) \geq \mu^*\left(\bigsqcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n}\mu^*(A_j)$$

where the last equality is obtained from (4). Since this is true for all n, for $n \to \infty$, we get

$$\mu^*\left(\bigsqcup_{j\geq 1} A_j\right) \geq \sum_{j\geq 1}\mu^*(A_j). \tag{6}$$

From (5) and (6) we obtain the required equality. Thus, we have shown that $\mu^*|_{\mathscr{M}}$ is a measure. We now show that it is a complete measure.

Let $A \subseteq \Omega$ such that $\mu^*(A) = 0$. Then, for any $E \subseteq \Omega$, $E \cap A \subseteq A$ and so by monotonicity, $\mu^*(E \cap A) \leq \mu^*(A) \implies \mu^*(E \cap A) = 0$. Also, $E \supseteq E \cap A^c \implies \mu^*(E) \geq \mu^*(E \cap A^c)$. Thus,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

which implies that $A \in \mathscr{M}$. Therefore, all sets of measure zero are in $\mathscr{M}$, and so $\mu^*|_{\mathscr{M}}$ is a complete measure.

QED

Definition 4.5. *(**Pre-measure.**) Let $\mathscr{A} \subseteq \mathscr{P}(\Omega)$ be an algebra. Then, $\mu_0 : \mathscr{A} \to [0,\infty]$ is said to be a pre-measure if:*

1. *$\mu_0(\varnothing) = 0$.*
2. *If $\{A_j\}_{j\geq 1}$ is a sequence of disjoint sets such that $\bigsqcup_{j\geq 1} A_j \in \mathscr{A}$, then $\mu_0\left(\bigsqcup_{j\geq 1} A_j\right) = \sum_{j\geq 1}\mu_0(A_j)$.*

Proposition 4.6. *Let Ω be a set and $\mathscr{A} \subseteq \mathscr{P}(\Omega)$ be an algebra. Define $\mu^* : \mathscr{P}(\Omega) \to [0, \infty]$ by*

$$\mu^*(A) := \inf \left\{ \sum_{j \geq 1} \mu_0(A_j) : A \subseteq \bigcup_{j \geq 1} A_j \text{ and } A_j \in \mathscr{A} \ \forall \ j \geq 1 \right\}$$

for all $A \in \mathscr{P}(\Omega)$. Further, let $\mathscr{M}$ be the collection of μ^*-measurable sets. Then,

1. *μ^* restricted to $\mathscr{A}$ is μ_0.*
2. *$\mathscr{A} \subseteq \mathscr{M}$.*

Proof. 1. Let $A \in \mathscr{A}$. We want to show that $\mu^*(A) = \mu_0(A)$. Set $A_1 = A$ and $A_j = \varnothing$ for all $j \geq 2$. Then, $\sum_{j \geq 1} \mu_0(A_j) = \mu_0(A)$ and so $\mu^*(A) \leq \mu_0(A)$.

We now show the reverse inequality. Let $\{A_j\}_{j \geq 1}$ be an arbitrary covering of A with $A_j \in \mathscr{A}$ for all $j \geq 1$. Then, $A \subseteq \bigcup_{j \geq 1} A_j$. Define $B_1 = A_1$ and $B_j = A_j \setminus \bigcup_{k=1}^{j-1} A_k$. Then, $B_i \cap B_j = \varnothing$ for all $i \neq j$. Clearly, $B_j \in \mathscr{A}$ as $\mathscr{A}$ is an algebra and therefore is closed under finite unions and set difference. Also, $A \subseteq \bigsqcup_{j \geq 1} B_j = \bigcup_{j \geq 1} A_j$. As μ_0 is a pre-measure, $\mu_0(A) \leq \sum_{j \geq 1} \mu_0(B_j)$. But by definition, $B_j \subseteq A_j$ and therefore $\mu_0(B_j) \leq \mu_0(A_j)$, implying that $\sum_{j \geq 1} \mu_0(B_j) \leq \sum_{j \geq 1} \mu_0(A_j)$. Thus, $\mu_0(A) \leq \sum_{j \geq 1} \mu_0(A_j)$. Since the covering $\{A_j\}_{j \geq 1}$ was arbitrary, $\mu_0(A) \leq \mu^*(A)$. Hence, $\mu^*(A) = \mu_0(A)$.

2. Let $A \in \mathscr{A}$. We want to show that $A \in \mathscr{M}$, that is,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for all $E \subseteq \Omega$.

Let $\varepsilon > 0$. There exists a covering $\{E_j\}_{j \geq 1}$ in $\mathscr{A}$ of $E \subseteq \Omega$ such that

$$\mu^*(E) \leq \sum_{j \geq 1} \mu_0(E_j) \leq \mu^*(E) + \varepsilon. \tag{7}$$

$E \subseteq \bigcup_{j \geq 1} E_j$. Let $F_1 = E_1$ and $F_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k$. Then, $\{F_j\}_{j \geq 1}$ is a sequence of disjoint sets with $F_j \in \mathscr{A}$ for all $j \geq 1$, since $\mathscr{A}$ is an algebra and so is closed under finite unions and set difference. Also, $\bigsqcup_{j \geq 1} F_j = \bigcup_{j \geq 1} E_j$ and so $\{F_j\}_{j \geq 1}$ is also a covering of E.

Now,

$$E \cap A \subseteq \left(\bigsqcup_{j \geq 1} F_j \right) \cap A = \bigsqcup_{j \geq 1} (F_j \cap A).$$

Since $A, F_j \in \mathscr{A}$, $F_j \cap A \in \mathscr{A}$. Further, observe that μ^* is an outer measure by Proposition 1.2. Using sub-additivity of μ^* yields

$$\mu^*(E \cap A) \leq \mu^* \left(\bigsqcup_{j \geq 1} (F_j \cap A) \right) \leq \sum_{j \geq 1} \mu_0(F_j \cap A).$$

Similarly,

$$\mu^*(E \cap A^c) \leq \sum_{j\geq 1} \mu_0(F_j \cap A^c).$$

Therefore,

$$\begin{aligned}\mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \sum_{j\geq 1} \mu_0(F_j \cap A) + \sum_{j\geq 1} \mu_0(F_j \cap A^c) \\ &= \sum_{j\geq 1} (\mu_0(F_j \cap A) + \mu_0(F_j \cap A^c)) \\ &= \sum_{j\geq 1} \mu_0(F_j).\end{aligned}$$

Since $F_j \subseteq E_j$, $\mu_0(F_j) \leq \mu_0(E_j)$. Hence, $\sum_{j\geq 1} \mu_0(F_j) \leq \sum_{j\geq 1} \mu_0(E_j)$, which gives

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \sum_{j\geq 1} \mu_0(E_j). \tag{8}$$

(7) and (8) imply

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E) + \varepsilon$$

but since this is true for all $\varepsilon > 0$, we obtain $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E)$, as required.

QED

Theorem 4.7. *Let $\mathscr{A} \subseteq \mathscr{P}(\Omega)$ be an algebra and let $\mu_0 : \mathscr{A} \to [0, \infty]$ be a pre-measure. Define $\mu^* : \mathscr{P}(\Omega) \to [0, \infty]$ by*

$$\mu^*(A) := \inf \left\{ \sum_{j\geq 1} \mu_0(A_j) : A \subseteq \bigcup_{j\geq 1} A_j \text{ and } A_j \in \mathscr{A} \ \forall\, j \geq 1 \right\}$$

for all $A \in \mathscr{P}(\Omega)$. Further, let $\mathscr{F}(\mathscr{A})$ be the σ-algebra generated by $\mathscr{A}$ and μ be the restriction of μ^* to $\mathscr{F}(\mathscr{A})$. Then:

1. *μ is a measure which extends the pre-measure μ_0 to $\mathscr{F}(\mathscr{A})$.*
2. *Let ν be any other measure which extends μ_0 to $\mathscr{F}(\mathscr{A})$. Then, $\nu(E) \leq \mu(E)$ for all $E \in \mathscr{F}(\mathscr{A})$, and if $\mu(E) < \infty$, then $\nu(E) = \mu(E)$. Further, if μ_0 is σ-finite, then μ is the unique extension of μ_0 to $\mathscr{F}(\mathscr{A})$.*

Proof. 1. We want to show that $\mu|_{\mathscr{A}} = \mu_0$. Let $\mathscr{M}$ be the collection of μ^*-measurable sets. By Theorem 2.2, $\mathscr{M}$ is a σ-algebra, and by Proposition 3.2, $\mathscr{A} \subseteq \mathscr{M}$. Therefore, $\mathscr{F}(\mathscr{A}) \subseteq \mathscr{M}$. By Proposition 3.2, we also have that $\mu^*|_{\mathscr{A}} = \mu_0$. Hence, $\mu|_{\mathscr{A}} = \mu_0$. Further, $\mu^*|_{\mathscr{M}}$ is a measure, as we showed in Theorem 2.2. This implies that $\mu^*|_{\mathscr{M}}$ possesses the property of countable additivity, and thus, $\mu^*|_{\mathscr{F}(\mathscr{A})}$ also possesses it, that is, μ is a measure.

2. We first show that if ν is any other measure which extends μ_0 to $\mathscr{F}(\mathscr{A})$, then $\nu(E) \leq \mu(E)$ for all $E \in \mathscr{F}(\mathscr{A})$. Let $\{E_j\}_{j\geq 1}$ be a covering of E in $\mathscr{A}$. Then, by the sub-additivity of ν, $\nu(E) \leq \sum_{j\geq 1} \nu(E_j)$. But since ν is an extension of μ_0 and $E_j \in \mathscr{A}$, $\nu(E_j) = \mu_0(E_j)$ and therefore, $\nu(E) \leq \sum_{j\geq 1} \mu_0(E_j)$. This is true for all coverings of E in $\mathscr{A}$ and therefore, $\nu(E) \leq \mu^*(E)$ by definition of μ^*. Now, $E \in \mathscr{F}(\mathscr{A})$ and $\mu = \mu^*|_{\mathscr{F}(\mathscr{A})}$ and thus, $\nu(E) \leq \mu(E) = \mu^*(E)$.

   Let $\mu(E) < \infty$. Suppose $E \subseteq \bigcup_{j\geq 1} E_j$ with $E_j \in \mathscr{A}$ for all $j \geq 1$. As usual, let $\{F_j\}_{j\geq 1}$ be the sequence of disjoint sets in $\mathscr{A}$ such that $\bigsqcup_{j=1}^n F_j = \bigcup_{j=1}^n E_j$. Then,

$$\begin{aligned}
\nu\left(\bigcup_{j\geq 1} E_j\right) &= \lim_{n\to\infty} \nu\left(\bigcup_{j=1}^n E_j\right) = \lim_{n\to\infty} \nu\left(\bigsqcup_{j=1}^n F_j\right) \\
&= \lim_{n\to\infty} \sum_{j=1}^n \mu_0(F_j) \\
&= \lim_{n\to\infty} \mu\left(\bigsqcup_{j=1}^n F_j\right) && \text{(as } \mu \text{ is also an extension of } \mu_0\text{)} \\
&= \lim_{n\to\infty} \mu\left(\bigcup_{j=1}^n E_j\right)
\end{aligned}$$

   Hence,

$$\nu\left(\bigcup_{j\geq 1} E_j\right) = \mu\left(\bigcup_{j\geq 1} E_j\right). \tag{9}$$

   As μ is a restriction of μ^* and by the way μ^* was defined, there exists a covering $\{E_j\}_{j\geq 1}$ of E in $\mathscr{A}$ such that $\mu(E) \leq \sum_{j\geq 1} \mu_0(E_j) \leq \mu(E) + \varepsilon$. Then, $\{F_j\}_{j\geq 1}$ is also a covering of E such that F_j are disjoint and $F_j \in \mathscr{A}$, with $\bigsqcup_{j\geq 1} F_j = \bigcup_{j\geq 1} E_j$. Since $F_j \subseteq E_j$, we have that $\mu_0(F_j) \leq \mu_0(E_j)$ and so,

$$\mu\left(\bigsqcup_{j\geq 1} F_j\right) = \sum_{j\geq 1} \mu(F_j) = \sum_{j\geq 1} \mu_0(F_j) \leq \sum_{j\geq 1} \mu_0(E_j) \leq \mu(E) + \varepsilon.$$

   That is, we just showed that for any $E \in \mathscr{F}(\mathscr{A})$, there exists a covering $\{F_j\}_{j\geq 1}$ in $\mathscr{A}$ such that if $F = \bigsqcup_{j\geq 1} F_j$, then $\mu(F) \leq \mu(E) + \varepsilon$. Also, since $E \subseteq F$, $\mu(F) = \mu(E) + \mu(F \setminus E)$, and so $\mu(F \setminus E) = \mu(F) - \mu(E) \leq \varepsilon$ (here we used the fact that $\mu(E) < \infty$). Now,

$$\begin{aligned}
\mu(E) \leq \mu(\bigcup_{j\geq 1} E_j) &= \nu(\bigcup_{j\geq 1} E_j) && \text{(from (9))} \\
&= \nu(F) \\
&= \nu(E) + \nu(F \setminus E) && \text{(as } \nu \text{ is a measure)} \\
&\leq \nu(E) + \mu(F \setminus E) \\
&\leq \nu(E) + \varepsilon.
\end{aligned}$$

This is true for all $\varepsilon > 0$ and thus $\mu(E) \leq \nu(E)$. Combining this with the fact that $\nu(E) \leq \mu(E)$ yields the equality $\nu(E) = \mu(E)$ if $\mu(E) < \infty$.

Finally, suppose that μ_0 is σ-finite. That is, $\Omega = \bigcup_{j\geq 1} E_j$ with $E_j \in \mathscr{A}$ and $\mu_0(E_j) < \infty$ for all $j \geq 1$. Then, for $F_1 = E_1$ and $F_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k$ for $j \geq 2$, $\Omega = \bigsqcup_{j\geq 1} F_j$, where F_j are disjoint and belong to $\mathscr{A}$. Also, note that since $F_j \subseteq E_j$, $\mu_0(F_j) < \infty$. Then, if ν is any other extension of μ_0 to $\mathscr{F}(\mathscr{A})$, for any $A \in \mathscr{F}(\mathscr{A})$,

$$\nu(A) = \nu\left(A \cap \Omega\right) = \nu\left(A \cap \bigsqcup_{j\geq 1} F_j\right) = \sum_{j\geq 1} \nu(A \cap F_j)$$

and since $A \cap F_j \subseteq F_j$, $\mu(A \cap F_j) < \infty$, implying that $\nu(A \cap F_j) = \mu(A \cap F_j)$, and thus,

$$\sum_{j\geq 1} \nu(A \cap F_j) = \sum_{j\geq 1} \mu(A \cap F_j) = \mu\left(A \cap \bigsqcup_{j\geq 1} F_j\right) = \mu(A).$$

Therefore, $\nu(A) = \mu(A)$, showing that μ is indeed the unique extension of μ_0 to $\mathscr{F}(\mathscr{A})$.
QED

5. Monotone Class Theorem

6. The Lebesgue Measure

7. Complete Measures

8. Approximation Theorems

9. Measurable Functions

10. Definition of the Integral

11. Properties of the Integral

In the discussion that follows:

- Let $(\Omega, \mathscr{F}, \mu)$ be a measure space.
- The notation $I(f)$ and $\int f d\mu$ is used interchangeably.

1. Let f, g be nonnegative integrable functions on Ω such that $g \leq f$. Then, $I(g) \leq I(f)$.

Proof. Let $f_n \uparrow f$ and $g_n \uparrow g$, where f_n and g_n are monotonically increasing sequences of simple functions. Define $h_n = \min\{f_n, g_n\}$. We first show that h_n is monotonically increasing, i.e., $h_n \leq h_{n+1}$ for all $n \in \mathbb{N}$.

Suppose $h_n = f_n$, i.e., $f_n \leq g_n$. If $h_{n+1} = f_{n+1}$, then $h_n \leq h_{n+1}$ since f_n is monotonically increasing. On the other hand, if $h_{n+1} = g_{n+1}$, then $g_{n+1} \geq g_n \geq f_n = h_n$. Similarly, if $h_n = g_n$, then also $h_n \leq h_{n+1}$, and so h_n is monotonically increasing.

Since $g \leq f$, for a large enough n, $g_n \leq f_n$. Hence, $\lim_{n\to\infty} h_n = \lim_{n\to\infty} g_n = g$ and so $h_n \uparrow g$. Also, by the way h_n is defined, $h_n \leq f_n$ for all $n \in \mathbb{N}$, and so $I(h_n) \leq I(f_n)$, since h_n and f_n are simple functions. Therefore, $I(g) = \lim_{n\to\infty} I(h_n) \leq \lim_{n\to\infty} I(f_n) = I(f)$. **QED**

2. Let $f : \Omega \to \mathbb{R}$ be an integrable function and $A \in \mathscr{F}$. Then, $\chi_A f$ is also integrable, and we denote $I(\chi_A f)$ as $\int_A f d\mu$.

Proof. Since $f^+ = \max\{0, f\}$ and since $\chi_A \in \{0, 1\}$, $(\chi_A f)^+ = \chi_A f^+ \leq f^+$. By Property 1, $I((\chi_A f)^+) \leq I(f^+) < \infty$ as f is integrable. Hence, $(\chi_A f)^+$ is integrable. Also, $f^- = \max\{0, -f\}$ and so, $(\chi_A f)^- = \chi_A f^- \leq f^-$. Once again applying Property 1, we get $I((\chi_A f)^-) \leq I(f^-) < \infty$, and thus, $(\chi_A f)^-$ is also integrable. Therefore, $\chi_A f$ is integrable. **QED**

3. Let $f, g : \Omega \to \mathbb{R}$ be nonnegative integrable functions. Then, $I(f + g) = I(f) + I(g)$ and $I(cf) = cI(f)$ for $c \geq 0$.

Proof. Let $f_n \uparrow f$ and $g_n \uparrow g$, where f_n, g_n are sequences of monotonically increasing, nonnegative simple functions. Then, $f_n + g_n \uparrow f + g$ and $f_n + g_n$ is a simple function for all $n \in \mathbb{N}$. Hence, $I(f+g) = \lim_{n\to\infty} I(f_n + g_n) = \lim_{n\to\infty}(I(f_n) + I(g_n))$, by the linearity of the integral over simple functions. Therefore, $I(f + g) = \lim_{n\to\infty} I(f_n) + \lim_{n\to\infty} I(g_n) = I(f) + I(g)$.

Also, $f_n \uparrow f$ implies that $cf_n \uparrow cf$ and since cf_n is also a monotonically increasing sequence of nonnegative simple functions (for $c \geq 0$), $I(cf) = \lim_{n\to\infty} I(cf_n) = \lim_{n\to\infty} cI(f_n) = cI(f)$, as desired. **QED**

4. Let $A, B \in \mathscr{F}$ be disjoint and $f : \Omega \to \mathbb{R}$ be integrable. Then, $\int_{A\cup B} f d\mu = \int_A f d\mu + \int_B f d\mu$.

Proof. We first show that if f is integrable over A and B, then it is also integrable over $A \cup B$. That is, $\chi_A f$ and $\chi_B f$ are integrable, and we want to show that $\chi_{A\cup B} f$ is also integrable.

$(\chi_{A\cup B} f)^+ = \chi_{A\cup B} f^+ = (\chi_A + \chi_B) f^+ = \chi_A f^+ + \chi_B f^+$. By Property 3, $I((\chi_{A\cup B} f)^+) = I(\chi_A f^+) + I(\chi_B f^+)$, and since f is integrable over A and B, $I(\chi_A f^+)$ and $I(\chi_B f^+)$ are both finite, and thus their sum is finite. Hence, $I((\chi_{A\cup B} f)^+)$ is finite. Similarly, $I((\chi_{A\cup B} f)^-)$ is also finite, and therefore $\chi_{A\cup B} f$ is integrable.

$\int_{A\cup B} f d\mu = \int \chi_{A\cup B} f d\mu$ and since A and B are disjoint,

$$\begin{aligned}
\int \chi_{A\cup B} f d\mu &= \int (\chi_{A\cup B} f)^+ d\mu - \int (\chi_{A\cup B} f)^- d\mu \\
&= \int \chi_A f^+ d\mu + \int \chi_B f^+ d\mu - \int \chi_A f^- d\mu - \int \chi_B f^- d\mu \qquad \text{(by Property 3)} \\
&= \left(\int \chi_A f^+ d\mu - \int \chi_A f^- d\mu \right) + \left(\int \chi_B f^+ d\mu - \int \chi_B f^- d\mu \right) \\
&= \int_A f d\mu + \int_B f d\mu.
\end{aligned}$$

QED

5. If $f : \Omega \to \mathbb{R}$ is integrable, then $|f| < \infty$ a.e.

Proof. f is integrable. We want to show that $|f| < \infty$ a.e., that is, if $E \in \mathscr{F}$ is such that $|f(x)| = \infty$ for all $x \in E$, then $\mu(E) = 0$.

Since $|f| = f^+ + f^-$, by Property 3 we have that $I(|f|) = I(f^+) + I(f^-)$, and since $I(f^+), I(f^-) < \infty$ as f is integrable, $I(|f|)$ is also finite and therefore $|f|$ is integrable.

We first note that

$$\int_{\{|f(x)|\geq n\}} |f| d\mu \leq \int |f| d\mu < \infty$$

and also that

$$\int_{\{|f(x)|\geq n\}} n d\mu \leq \int_{\{|f(x)|\geq n\}} |f| d\mu.$$

Hence,

$$n\mu(\{|f(x)| \geq n\}) \leq \int_{\{|f(x)|\geq n\}} |f| d\mu,$$

i.e.,

$$\mu(\{|f(x)| \geq n\}) \leq \frac{1}{n} \int_{\{|f(x)|\geq n\}} |f| d\mu.$$

Now, define $A_n := \{x : |f(x)| \geq n\}$. Since f is integrable, $A_n \in \mathscr{F}$. Clearly, $A_{n+1} \subseteq A_n$. Define $A := \bigcap_{n\geq 1} A_n$. Then, $A \in \mathscr{F}$. Define $E := \{x : |f(x)| = \infty\}$. We now show that $E = A$.

Let $x \in A$. Then, $x \in A_n$ and therefore $|f(x)| \geq n$ for all $n \in \mathbb{Z}$. Hence, $|f(x)| = \infty$ and so $x \in E$. Thus, $A \subseteq E$. Now suppose $x \in E$. Then, $|f(x)| = \infty$ and so $|f(x)| \geq n$ for all $n \in \mathbb{N}$, implying that $x \in A$. Thus, $E \subseteq A$ and therefore $A = E$.

Since $\mu(A_1)$ is finite, we can apply continuity from above to obtain

$$\mu(E) = \lim_{n\to\infty} \mu(A_n) \leq \lim_{n\to\infty} \frac{1}{n} \int_{A_n} |f| d\mu \leq \lim_{n\to\infty} \frac{1}{n} \int |f| d\mu = 0$$

as desired. **QED**

6. If $f, g : \Omega \to \mathbb{R}$ are nonnegative and integrable, then $\int (f-g)d\mu = \int f d\mu - \int g d\mu$.

Proof. Define $h := f - g$. We assume that h is well-defined, that is, f and g are not $+\infty$ simultaneously. Then, $h = h^+ - h^- = f - g$. If h is finite, then $h^+ + g = h^- + f$. If $h = +\infty$, then $f = +\infty$, g is finite, $h^+ = +\infty$, and h^- is finite. Therefore, $h^+ + g = +\infty = h^- + f$. Similarly, $h^+ + g = h^- + f$ holds even when $h = -\infty$. Hence, $\int (h^+ + g)d\mu = \int (h^- + f)d\mu$. Applying Property 3 gives $\int h^+ d\mu + \int g d\mu = \int h^- d\mu + \int f d\mu$, i.e., $\int h^+ d\mu - \int h^- d\mu = \int f d\mu - \int g d\mu$, and since $\int h d\mu = \int h^+ d\mu - \int h^- d\mu$, we have the desired result. **QED**

7. If $f, g : \Omega \to \mathbb{R}$ are integrable, then $f + g$ is also integrable and $\int (f+g)d\mu = \int f d\mu + \int g d\mu$.

Proof. We first show that $f+g$ is integrable. $(f+g)^+ \leq f^+ + g^+$ and therefore, by Properties 1 and 3, $I((f+g)^+) \leq I(f^+) + I(g^+) < \infty$ since $I(f^+), I(f^+) < \infty$, as f, g are integrable. Similarly, $I((f+g)^-) < \infty$ and so $f + g$ is integrable.

Now, $f + g = (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-)$. By Property 5,

$$\begin{aligned}
\int (f+g)d\mu &= \int (f^+ + g^+)d\mu - \int (f^- + g^-)d\mu \\
&= \left(\int f^+ d\mu + \int g^+ d\mu\right) - \left(\int f^- d\mu + \int g^- d\mu\right) \qquad \text{(by Property 3)} \\
&= \left(\int f^+ d\mu - \int f^- d\mu\right) + \left(\int g^+ d\mu - \int g^- d\mu\right) \\
&= \int f d\mu + \int g d\mu.
\end{aligned}$$

QED

8. If $f : \Omega \to \mathbb{R}$ is integrable, then $|\int f d\mu| \leq \int |f| d\mu$.

Proof. We have,

$$\begin{aligned}
\left|\int f d\mu\right| &= \left|\int f^+ d\mu - \int f^- d\mu\right| \\
&\leq \int f^+ d\mu + \int f^- d\mu \\
&= \int (f^+ + f^-)d\mu = \int |f| d\mu.
\end{aligned}$$

QED

9. Let $f : \Omega \to \mathbb{R}$ be integrable and $c \in \mathbb{R}$. Then, cf is integrable and $\int cf d\mu = c \int f d\mu$.

Proof. $\int |cf|d\mu = \int |c||f|d\mu = |c| \int |f|d\mu$, where the last equality is true by Property 3. $\int |f|d\mu = \int (f^+ + f^-)d\mu = \int f^+ d\mu + \int f^- d\mu < \infty$ as f is integrable, and so $\int |cf|d\mu < \infty$. Since $\int (cf)^+ d\mu$ and $\int (cf)^- d\mu$ are both bounded by $\int |cf|d\mu$, they are also finite and hence cf is integrable.

If $c = 0$, the statement is trivial.

If $c > 0$,

$$\begin{aligned}\int cf d\mu &= \int (cf)^+ d\mu - \int (cf)^- d\mu \\ &= \int cf^+ d\mu - \int cf^- d\mu \\ &= c\left(\int f^+ d\mu - \int f^- d\mu\right) \\ &= c \int f d\mu.\end{aligned}$$

Further, if $c < 0$,

$$\begin{aligned}\int cf d\mu &= \int (cf)^+ d\mu - \int (cf)^- d\mu \\ &= \int (-c) f^- d\mu - \int (-c) f^+ d\mu \\ &= c \int f d\mu.\end{aligned}$$

QED

10. If $f \geq 0$, then $\int f d\mu \geq 0$.

Proof.

$$\begin{aligned}\int f d\mu &= \int f^+ d\mu - \int f^- d\mu \\ &= \int f^+ d\mu - 0 \geq 0.\end{aligned}$$

QED

11. If $f \geq g$, then $\int f d\mu \geq \int g d\mu$.

Proof. Since $f \geq g$, $f - g \geq 0$. By Property 10, $\int (f - g)d\mu \geq 0$, and by linearity, $\int f d\mu - \int g d\mu \geq 0$, i.e., $\int f d\mu \geq \int g d\mu$. **QED**

12. If $f \geq 0$ and $\int f d\mu = 0$, then $f = 0$ a.e.

Proof. Define $E_n := \{x : f(x) \geq \frac{1}{n}\}$. Let $E = \bigcup_{n \geq 1} E_n$ and $F = \{x : f(x) > 0\}$. We first show that $E = F$. Let $x \in E$. Then, for some $n \geq 1$, $x \in E_n$, i.e., $f(x) \geq \frac{1}{n}$, and since $n \in \mathbb{N}$, $f(x) > 0$. Hence, $x \in F$ and $E \subseteq F$. Also, if $x \in F$, then $f(x) > 0$, and so there

exists $n \in \mathbb{N}$ such that $f(x) \geq \frac{1}{n}$, i.e., $x \in E_n$. Therefore, $F \subseteq E$, showing that $E = F$. Further, $E_n \subseteq E_{n+1}$ and so $E_n \uparrow E$.

Now,

$$\int_{E_n} \frac{1}{n} d\mu \leq \int_{E_n} f d\mu \leq \int f d\mu = 0$$

and so $\int_{E_n} \frac{1}{n} d\mu = 0$. Hence, $\frac{1}{n} \int \chi_{E_n} d\mu = 0$, i.e., $\mu(E_n) = 0$, for all $n \in \mathbb{N}$. By the subadditivity of μ, $\mu(E) \leq \sum_{n \geq 1} \mu(E_n) = 0$. Since $E^c = \{x : f(x) = 0\}$ (as $f \geq 0$), and since $\mu(E) = 0$, $f = 0$ a.e. **QED**

13. If $f = g$ a.e., then $\int f d\mu = \int g d\mu$.

Proof. $f = g$ a.e., i.e., $f(x) = g(x)$ for all $x \in E$ and $\mu(E^c) = 0$. $\int f d\mu = \int (\chi_E + \chi_{E^c}) f d\mu = \int \chi_E f d\mu + \int \chi_{E^c} f d\mu$. Since $\mu(E^c) = 0$, $\int \chi_{E^c} f d\mu = 0$, and so $\int f d\mu = \int \chi_E f d\mu = \int \chi_E g d\mu$, as $f = g$ over E. Also, $\int \chi_{E^c} g d\mu = 0$ and so $\int \chi_E g d\mu = \int \chi_E g d\mu + \int \chi_{E^c} g d\mu = \int g d\mu$. Therefore, $\int f d\mu = \int g d\mu$. **QED**

14. Let $h : \Omega \to \mathbb{R}$ be measurable and $|h| \leq f$, where f is integrable. Then, h is integrable.

Proof. Since $|h| \leq f$, $h^+ + h^- \leq f$. By Property 11, $I(h^+ + h^-) \leq I(f)$, and by Property 3, $I(h^+) + I(h^-) \leq I(f)$. Thus, $I(h^+), I(h^-) \leq I(f)$, and since f is integrable, $I(f)$ is finite. Therefore, both $I(h^+)$ and $I(h^-)$ are finite, and so h is integrable. **QED**

15. Let $f : \Omega \to \mathbb{R}$ be a measurable function which is bounded on E, where $\mu(E) < \infty$, and $f = 0$ on E^c. Then, f is integrable.

Proof. We want to show that $I(f^+)$ and $I(f^-)$ are finite, but since both of them are bounded by $I(|f|)$, it suffices to show that $I(|f|)$ is finite.

Since f is bounded on E, $|f| < c$ for some $c \in \mathbb{R}$. Hence, $\int_E |f| d\mu \leq \int_E c d\mu = \int \chi_E c d\mu = c\mu(E) < \infty$, since $\chi_E c$ is a simple function. Further, $\int_{E^c} |f| d\mu = 0$, and so $\int |f| d\mu = \int_E |f| d\mu + \int_{E^c} |f| d\mu < \infty$, as required. **QED**

12. Theorems on the Convergence of Integrals

13. Classes of Subsets in Product Spaces

14. Product Measures

15. Fubini's Theorem

16. Radon-Nikodym Theorem

17. Types of Convergence

Let $(\Omega, \mathscr{F}, \mu)$ be a σ-finite measure space and let $\mathscr{F}$ be μ-complete. **Recall:** If $f, g : \Omega \to \bar{\mathbb{R}}$ are functions such that f is $\mathscr{F}$-measurable and $g = f$ a.e., then g is also $\mathscr{F}$-measurable.

We define a relation $\sim$ as follows: $f \sim g$ if $f = g$ a.e. Clearly, $\sim$ is an equivalence relation. Let M be the set of all $\mathscr{F}$-measurable functions and $\mathscr{M} := M/\sim$, the collection of equivalence classes induced by $\sim$.

Outline. We will study the following types of convergence:

1. Almost-everywhere convergence
2. Almost-everywhere uniform convergence
3. Almost uniform convergence

Almost-everywhere convergence

We first recall the definition of *pointwise convergence*:

Definition 17.1. *Let $f, f_n : E \to \bar{\mathbb{R}}$ such that for all $x \in E$, $f_n(x) \to f(x)$. Then $(f_n)_{n\geq 1}$ **converges pointwise** to f.*

Definition 17.2. *We say f_n **converges to f almost everywhere** if there exists $E \in \mathscr{F}$ such that $f_n \to f$ pointwise on E and $\mu(E^c) = 0$.*

Lemma 17.3. *Let $f_n \to f$ a.e. and $f_n \sim g_n$ for all n. Further, let $f \sim g$. Then, $g_n \to g$ a.e.*

Proof. Since $f_n \to f$ a.e., there exists $A \in \mathscr{F}$ such that $\mu(A^c) = 0$ and $f_n \to f$ pointwise on A. Also, since $f \sim g$, there exists $B \in \mathscr{F}$ such that $\mu(B^c) = 0$ and $f(x) = g(x)$ for all $x \in B$.

Now, $f_n \sim g_n$ and so, for every $n \geq 1$, there is some $E_n \in \mathscr{F}$ with $\mu(E_n^c) = 0$ and $f_n(x) = g_n(x)$ for all $x \in E_n$.

Define

$$E := A \cap B \cap \bigcap_{n\geq 1} E_n.$$

Then,

$$E^c = A^c \cup B^c \cup \bigcup_{n\geq 1} E_n^c$$

and so,

$$\mu(E^c) \leq \mu(A^c) + \mu(B^c) + \sum_{n\geq 1} \mu(E_n^c) = 0,$$

i.e., $\mu(E^c) = 0$. Further, since $f_n(x) \to f(x)$ for all $x \in A$ and since $E \subseteq A$, we have that $f_n(x) \to f(x)$ for all $x \in E$. But for any $x \in E$, $x \in E_n$ for all $n \geq 1$, and so $f_n(x) = g_n(x)$ for all $n \geq 1$. Hence, $g_n(x) \to f(x)$ for all $x \in E$. Lastly we observe that $E \subseteq B$ and so $f(x) = g(x)$ for all $x \in E$. Thus, $g_n(x) \to g(x)$ for all $x \in E$, and since $\mu(E^c) = 0$, we have $g_n \to g$ a.e. **QED**

Lemma 17.4. *If $f_n \to f$ a.e. and $f_n \to g$ a.e., then $f = g$ a.e.*

Proof. $f_n \to f$ and so for some $E \in \mathscr{F}$, $\mu(E^c) = 0$ and $f_n \to f$ pointwise on E. Similarly, $f_n \to g$ pointwise on some $F \in \mathscr{F}$ with $\mu(F^c) = 0$. Define $H := E \cap F$. For all $x \in H$, $f_n(x) \to f(x)$ and $f_n(x) \to g(x)$. By the uniqueness of the limit of a sequence, we have $f(x) = g(x)$ for all $x \in H$. Also, $\mu(H^c) = \mu(E^c \cup F^c) \leq \mu(E^c) + \mu(F^c) = 0$ and so $\mu(H^c) = 0$. Hence, $f = g$ a.e. **QED**

Example 17.5. Let $\Omega = [0, 1]$, and consider the measure space $(\Omega, \mathscr{L}, \lambda)$, where $\mathscr{L}$ is the Lebesgue σ-algebra on Ω and λ is the Lebesgue measure. Define $f_n(x) := x^n$. Then, $f_n(x) \to 0$ for all $x \in [0, 1)$ but $f_n(1)$ does not converge to 0. Hence, $(f_n)_{n\geq 1}$ does not converge pointwise to 0 on $[0, 1]$, but $f_n \to 0$ λ-a.e.

Almost-everywhere uniform convergence

We recall the definition of *uniform convergence*:

Definition 17.6. *Let $f, f_n : E \to \bar{\mathbb{R}}$. We say that f_n **converges to f uniformly** if $\forall \epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $|f_n(x) - f(x)| < \epsilon$ for all $x \in E$; or, equivalently, for all $n \geq n_0$,*

$$\sup_{x\in E} |f_n(x) - f(x)| < \epsilon \quad \forall x \in E.$$

Definition 17.7. *Let $f, f_n : \Omega \to \bar{\mathbb{R}}$. Then, f_n **converges to f uniformly almost everywhere** if there exists some $E \in \mathscr{F}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformly on E.*

Definition 17.8. *Let $f : \Omega \to \bar{\mathbb{R}}$ be a function and suppose there exists some $a > 0$ such that $\mu\{x : |f(x)| > a\} = 0$. We define the **essential supremum** of f by*

$$\text{ess sup } |f| = \inf\{a > 0 : \mu\{x : |f(x)| > a\} = 0\}.$$

Clearly, if $\mu\{x : |f(x)| > a\} = 0$ and $b > a$ then $\mu\{x : |f(x)| > b\} = 0$. Also, ess sup$|f| \leq \sup |f|$.